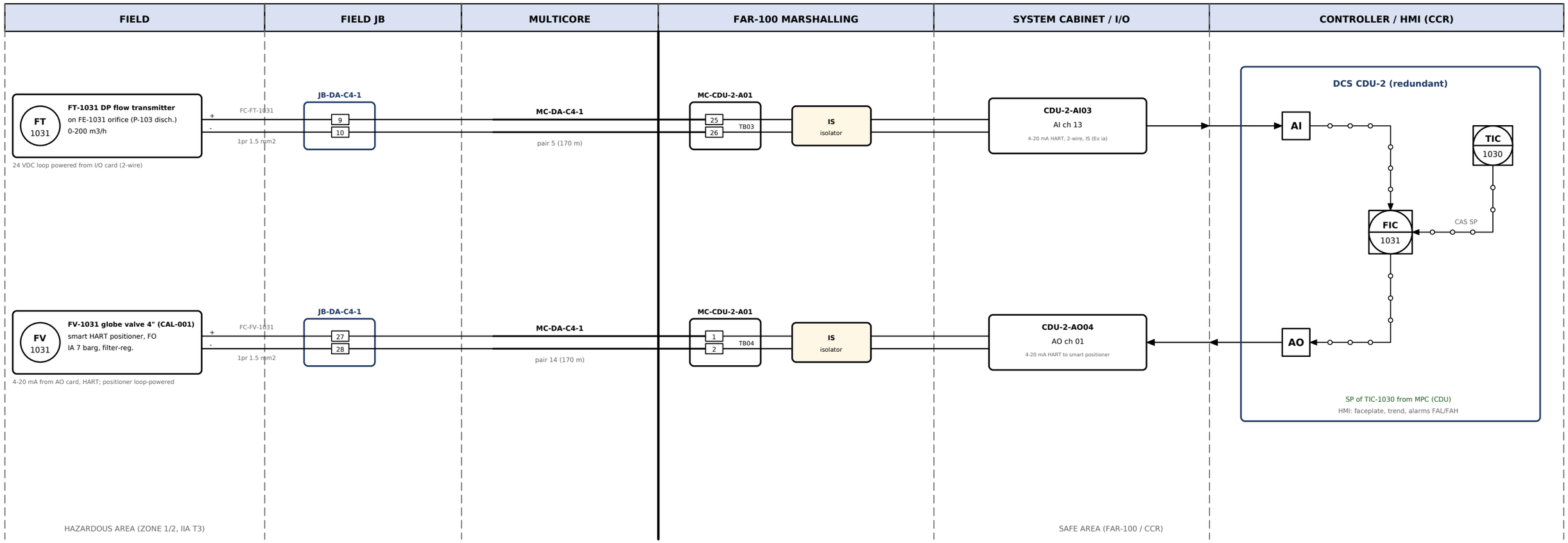


NOTES

1. Cable FC- = field (branch) cable 1 pair 1.5 mm2 screened, IS blue sheath; multicore 12/24 pair, armoured, overall + individual screens; screens earthed at FAR-100 IS earth bar only.
2. Terminal numbers, JB, multicore, cabinet and card/channel assignments from CFU-000-IC-IOL-001 (data/io_list.json) - final at detailed design.
3. Positioner fail action: air failure -> valve OPEN (FO, reflux to column top on air loss).
4. FIC-1031 in CAS from TIC-1030; on TIC bad PV the FIC sheds to AUTO at last SP (bumpless).
5. Power: AI/AO cards 24 VDC from redundant cabinet PSUs fed by UPS-101A/B (UDB-101A/B).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

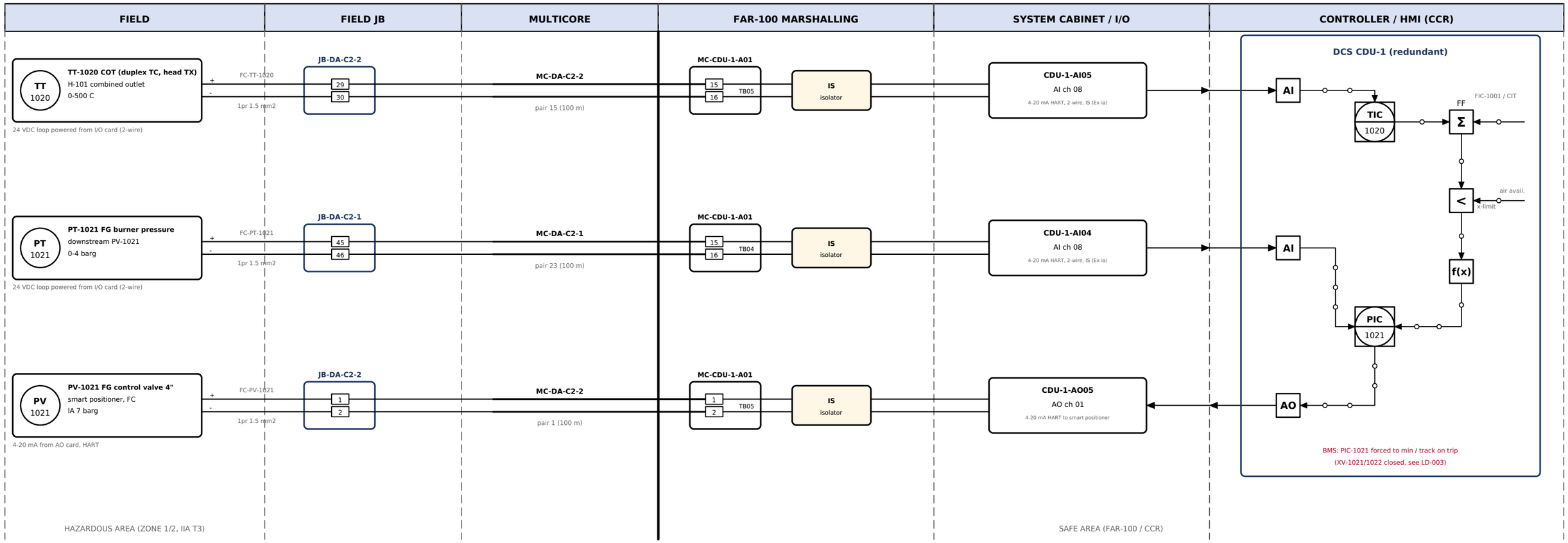
LOOP DIAGRAM

FIC-1031 C-101 REFLUX FLOW (CASCADE FROM TIC-1030)

DISCIPLINE	STAGE	SCALE	SHEET
INSTRUMENTATION	FEED	NTS	1 OF 3

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER	REV
CFU-100-IC-LD-001	**A**



NOTES

1. TT-1020 (BPCS) is independent of the SIS transmitters TT-1020A/B/C (SIF-106); separate thermowells.
2. Cascade: TIC-1020 output + feed-forward (charge x dT, lead-lag) -> fuel demand -> low select with air available (cross-limiting) -> burner curve f(x) -> PIC-1021 SP. See CFU-100-IC-CSD-001.
3. PV-1021 fail-closed (FC) on air failure; min-fire mechanical stop not used (min-fire in DCS logic).
4. Terminal / cable data from CFU-000-IC-IOL-001 (data/io_list.json).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

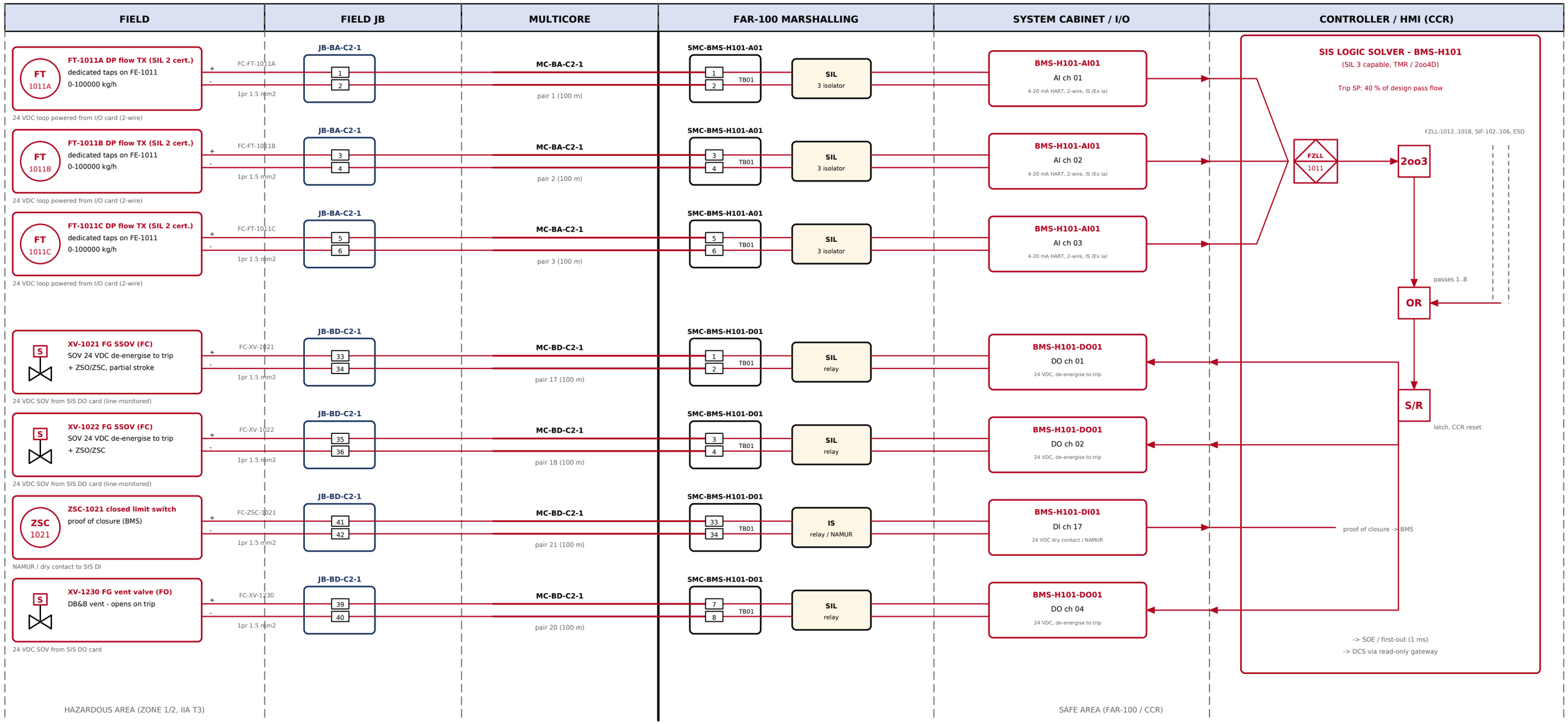
LOOP DIAGRAM

TIC-1020 / PIC-1021 H-101 COT -> FUEL GAS PRESSURE CASCADE

DISCIPLINE	STAGE	SCALE	SHEET
INSTRUMENTATION	FEED	NTS	2 OF 3

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER	REV
CFU-100-IC-LD-002	**A**



NOTES

1. SIL 2 per LOPA (CFU-000-IC-RPT-002). 2oo3 voting with 2oo2 on one transmitter in bypass / fault; deviation alarm between A/B/C (5 %).
2. Final elements: XV-1021 and XV-1022 in series (1oo2) with vent XV-1230 (double block & bleed); also pilot XV-1026 (SIF-101 / BMS). Proof-test interval 12 months (partial stroke 3 months).
3. Field cables and JBs for SIS are segregated (red sheath / labels); SIS multicores separate from DCS.
4. Terminal / cable data from CFU-000-IC-IOL-001 (data/io_list.json).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

LOOP DIAGRAM

SIF-101 H-101 PASS 1 LOW-LOW FLOW (2OO3) -> FUEL SSOVS

DISCIPLINE	STAGE	SCALE	SHEET
INSTRUMENTATION	FEED	NTS	3 OF 3

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER	REV
CFU-100-IC-LD-003	**A**